ISyE6669 Deterministic Optimization
Homework 11

Fall 2025

Dantzig-Wolfe decomposition

1. Consider the following linear programming problem:

$$
\begin{array}{lllllllllll}
\max & x_{11} & & & & & +x_{22} & & & \\
\text{s.t.} & x_{11} & & & +x_{14} & +x_{21} & -x_{22} & & & \leq 20 \\
 & x_{11} & +x_{12} & +x_{13} & +x_{14} & & & & & = 16 \\
 & & & & & x_{21} & +x_{22} & +x_{23} & +x_{24} & = 16 \\
 & x_{11} & & & & +x_{21} & & & & = 8 \\
 & & x_{12} & & & & +x_{22} & & & = 8 \\
 & & & x_{13} & & & & +x_{23} & & = 8 \\
 & & & & x_{14} & & & & +x_{24} & = 8 \\
 & & & & & x_{ij} \geq 0, & \multicolumn{4}{l}{\text{for all } i = 1, 2,\ j = 1, 2, 3, 4.}
\end{array}
$$

We will solve this problem using the Dantzig-Wolfe decomposition. Consider the first constraint $x_{11}+x_{14}+x_{21}-x_{22} \leq 20$ as the "complicating" constraint (i.e. the $\boldsymbol{Dx} \leq \boldsymbol{b}$ constraint) and consider the remaining constraints including nonnegativity constraints as the "easy" constraints, which define a polyhedron P. That is, P is defined by the above five equality constraints and the nonnegativity constraints.

In the following, we will guide you through a few steps to solve the problem. These steps will help you review some important topics such as reduced costs, the simplex method, LP duality (including complementary slackness), and column generation.

(a) First, argue that the polyhedron P defined above is bounded.

Solution: As all 8 variables are constrained to be non-negative and involved in an equality constraint with a RHS value of 16 or less, they are all bounded above by 16, meaning that P exists within a 8-dimensional hypercube with side length 16 (i.e., P is bounded).

An alternative solution is as follows. First, note that all variables are constrained to the positive orthant, so P cannot contain a line and any extreme ray of P must be $\geq \mathbf{0}$. Suppose for contradiction that P contains an extreme ray, i.e., there exists a direction $\boldsymbol{d}$ such that we can move indefinitely away from some fixed $\boldsymbol{x}$ in P. Then $\boldsymbol{d}$ must satisfy $\boldsymbol{A}(\boldsymbol{x} + \theta \boldsymbol{d}) = \boldsymbol{b}$ for all $\boldsymbol{x}$ in P, for all $\theta \geq 0$, so $\boldsymbol{d}$ must satisfy $\boldsymbol{Ad} = \mathbf{0}, \boldsymbol{d} \geq 0$. Thus it is not possible to have a solution other than $\boldsymbol{d} = \mathbf{0}$, so there are no extreme rays and therefore P must be bounded.

Note: An argument that there are are no extreme rays must include valid reasoning; otherwise, it is insufficient.

(b) Since P is bounded, we can use the extreme point representation for P. The Dantzig-Wolfe master problem can be written as:

$$\begin{aligned} \max \quad & \sum_{i=1}^{N} \lambda_i \left(\boldsymbol{c}^T \boldsymbol{x}^i \right) \\ \text{s.t.} \quad & \sum_{i=1}^{N} \lambda_i \left(\boldsymbol{D} \boldsymbol{x}^i \right) \leq \boldsymbol{b} \\ & \sum_{i=1}^{N} \lambda_i = 1, \\ & \lambda_i \geq 0, \quad \forall i = 1, \ldots, N. \end{aligned}$$

Specify $\boldsymbol{c}$, $\boldsymbol{D}$, and $\boldsymbol{b}$ for this problem. Note that each extreme points $\boldsymbol{x}^i$ of the polyhedron P has 8 components, i.e. $\boldsymbol{x}^i = (x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24})$.

Solution:

$$\boldsymbol{c} = (1, 0, 0, 0, 0, 1, 0, 0), \ \boldsymbol{D} = (1, 0, 0, 1, 1, -1, 0, 0), \ \boldsymbol{b} = 20.$$

(c) You are given the following two extreme points of the polyhedron P:

$$\boldsymbol{x}^1 = (x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}) = (8, 0, 8, 0, 0, 8, 0, 8),$$

and

$$\boldsymbol{x}^2 = (x_{11}, x_{12}, x_{13}, x_{14}, x_{21}, x_{22}, x_{23}, x_{24}) = (0, 0, 8, 8, 8, 8, 0, 0).$$

Construct the restricted master problem using these two extreme points. Use variables λ_1 and λ_2 for the restricted master problem.

Solution: The restricted master problem can be written as

$$\begin{aligned} \max_{\lambda_1, \lambda_2} \quad & 16\lambda_1 + 8\lambda_2 \\ \text{s.t.} \quad & 8\lambda_2 \leq 20, \\ & \lambda_1 + \lambda_2 = 1, \\ & \lambda_1, \lambda_2 \geq 0. \end{aligned}$$

(d) Find the optimal solution of the restricted master problem, which can be solved by hand.

Solution: By inspection or plotting (see Figure 1), optimal solution is $(\lambda_1, \lambda_2) = (1, 0)$.

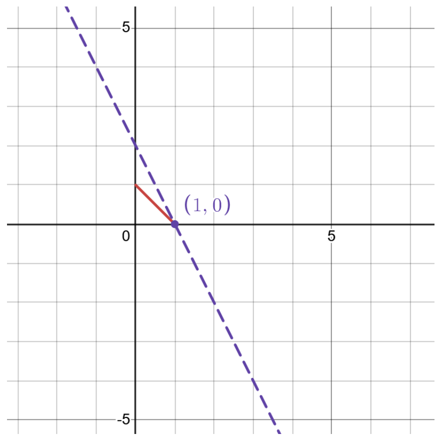

Figure 1: RMP

(e) Find the optimal dual variables in *two ways*. First, find the basis matrix for the optimal solution of the restricted master problem and compute the dual variables $\hat{\boldsymbol{y}}$ using $\hat{\boldsymbol{y}}^T = \boldsymbol{c}_B^T \boldsymbol{B}^{-1}$. The second way: Form the dual problem of the restricted master problem, and compute the optimal dual variables using Complementary Slackness. The two ways should give the same answer.

Solution:

Method 1: Form part (d) we know that solution of RMP is ($\lambda_1 = 1, \lambda_2 = 0$). To find basis matrix we need to reformulate problem in standard form:

$$\begin{aligned} \max_{\lambda_1,\lambda_2} \quad & 16\lambda_1 + 8\lambda_2 \\ \text{s.t.} \quad & 8\lambda_2 + s = 20, \\ & \lambda_1 + \lambda_2 = 1, \\ & \lambda_1, \lambda_2, s \geq 0. \end{aligned}$$

Solution of the problem in standard form is ($\lambda_1 = 1, \lambda_2 = 0, s = 20$), hence basic variables are λ_1 and s and the basis matrix for the optimal solution of the restricted master problem is given as $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Its inverse is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Cost soefficients corresponding to basic variables are $c_B^{\mathrm{T}} = \begin{bmatrix} 16 & 0 \end{bmatrix}$. The dual variables, $\hat{\mathbf{y}}$, are thus

$$\hat{\mathbf{y}}^{\mathrm{T}} = \mathbf{c}_B^{\mathrm{T}} \mathbf{B}^{-1} = \begin{bmatrix} 16 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 16 \end{bmatrix} = \begin{bmatrix} y & r \end{bmatrix}. \tag{1}$$

Method 2: Dual problem of the RMP is:

$$\begin{aligned}
\min_{y,r} \quad & 20y + r \\
\text{s.t.} \quad & r \geq 16 \\
& 8y + r \geq 8 \\
& y \geq 0.
\end{aligned}$$

To find the optimal solution to the dual problem, we will use complementary slackness:

i. Primal Complementary Slackness: Since primal constraint $8\lambda_2 \leq 20$ is strict at the primal optimal solution $\lambda_2 = 0$ it means that corresponding dual variable $y = 0$.

ii. Dual Complementary Slackness: Since primal variable $\lambda_1 = 1 \neq 0$ it means that dual constraint $r \geq 16$ must be active and hence $r = 16$.

As noted in the problem statements, both methods arrive at the same solution.

(f) Using the dual variables computed in the previous part, formulate the subproblem that maximizes the reduced cost of the restricted master problem.

Solution: The subproblem for maximum reduced cost can be written as

$$\begin{aligned}
Z = \max_{\boldsymbol{x}} \quad & \boldsymbol{c}^\top \boldsymbol{x} - \hat{y}^\top (\boldsymbol{D}\boldsymbol{x}) - \hat{r} \\
\text{s.t.} \quad & \boldsymbol{x} \in P, \\
= \max_{\boldsymbol{x}} \quad & x_{11} + x_{22} - 16 \\
\text{s.t.} \quad & x_{11} + x_{12} + x_{13} + x_{14} = 16 \\
& x_{21} + x_{22} + x_{23} + x_{24} = 16 \\
& x_{11} + x_{21} = 8 \\
& x_{12} + x_{22} = 8 \\
& x_{13} + x_{23} = 8 \\
& x_{14} + x_{24} = 8 \\
& x_{ij} \geq 0, \quad \text{for all } i = 1, 2,\ j = 1, 2, 3, 4.
\end{aligned}$$

(g) If the maximum reduced cost is Z, when should we terminate the Dantzig-Wolfe decomposition algorithm?

Solution: Because this is a maximization problem, if Z is nonpositive, we can safely terminate the algorithm.

(h) The polytope P has an interesting interpretation. Think about x_{ij} as the amount of product shipped from warehouse i to city j. Use this to interpret the six equality constraint as flow conservation constraints. This problem is about scheduling shipment across a network of warehouses and cities.

Solution: Following this interpretation, the first two equality constraints mean that the total amount of product shipped from warehouse 1 and 2 are both 16 units, while the total amount of product shipped to each city is 8 units.

2. Consider the following linear program:

$$\begin{aligned} \min \quad & 4x_1 + 3x_2 + x_3 + 2x_4 \\ \text{s.t.} \quad & 2x_1 - x_2 + x_3 + 3x_4 \leq 6 \\ & 2x_1 + x_2 \leq 4 \\ & -x_1 + x_2 \leq 1 \\ & x_2 \geq 0 \\ & 2x_3 + x_4 \leq 5 \\ & x_3 - x_4 \leq 1 \\ & x_3, x_4 \geq 0 \end{aligned}$$

Denote the polyhedron for variables x_1 and x_2 defined by constraints $2x_1 + x_2 \leq 4,\ -x_1 + x_2 \leq 1$, and $x_2 \geq 0$ as P_1 and denote the polyhedron for x_3 and x_4 defined by constraints $2x_3 + x_4 \leq 5,\ x_3 - x_4 \leq 1$ and $x_3, x_4 \geq 0$ as P_2. Use two extreme points of P_1 :

$$(x_1^1, x_2^1) = (1, 2)$$
$$(x_1^2, x_2^2) = (2, 0),$$

and two extreme points of P_2 :

$$(x_3^1, x_4^1) = (0, 5)$$
$$(x_3^2, x_4^2) = (1, 0).$$

Construct the restricted master problem using these extreme points. Use variables α_1 and α_2 for P_1 and β_1 and β_2 for P_2 in the restricted master problem.

Solution: Every point from the feasible set can be written as:

$$(x_1, x_2, x_3, x_4) = (x_1, x_2, 0, 0) + (0, 0, x_3, x_4)$$

such that $2x_1 - x_2 + x_3 + 3x_4 \leq 6$ and $(x_1, x_2) \in P_1$ and $(x_3, x_4) \in P_2$. By using extreme point representation for points in P_1 we get:

$$\begin{aligned} & (x_1, x_2) = \alpha_1(1, 2) + \alpha_2(2, 0) = (\alpha_1 + 2\alpha_2, 2\alpha_1) \\ & \alpha_1 + \alpha_2 = 1 \\ & \alpha_1, \alpha_2 \geq 0 \end{aligned}$$

By using extreme point representation for points in P_2 we get:

$$\begin{aligned} & (x_3, x_4) = \beta_1(0, 5) + \beta_2(1, 0) = (\beta_2, 5\beta_1) \\ & \beta_1 + \beta_2 = 1 \\ & \beta_1, \beta_2 \geq 0. \end{aligned}$$

By substituting

$$(x_1, x_2, x_3, x_4) = (\alpha_1 + 2\alpha_2, 2\alpha_1, \beta_2, 5\beta_1)$$

into the objective function and complicating constraint we get:

$$\begin{aligned} 4x_1 + 3x_2 + x_3 + 2x_4 &= 4(\alpha_1 + 2\alpha_2) + 3(2\alpha_1) + \beta_2 + 2(5\beta_1) = \\ &= 10\alpha_1 + 8\alpha_2 + 10\beta_1 + \beta_2 \end{aligned}$$

and

$$\begin{aligned} 2x_1 - x_2 + x_3 + 3x_4 \leq 6 &\quad \Leftrightarrow \\ 2(\alpha_1 + 2\alpha_2) - 2\alpha_1 + \beta_2 + 3(5\beta_1) \leq 6 &\quad \Leftrightarrow \\ 4\alpha_2 + 15\beta_1 + \beta_2 \leq 6 & \end{aligned}$$

Finally, we get restricted master problem:

$$\begin{array}{cc} \min & 10\alpha_1 + 8\alpha_2 + 10\beta_1 + \beta_2 \\ \text{s.t.} & 4\alpha_2 + 15\beta_1 + \beta_2 \leq 6 \\ & \alpha_1 + \alpha_2 = 1 \\ & \beta_1 + \beta_2 = 1 \\ & \alpha_1, \alpha_2, \beta_1, \beta_2 \geq 0 \end{array}$$

Image compression and SVD

3. Singular Value Decomposition (SVD) is a very useful tool in data analysis. In this exercise, you will use SVD to compress image. Implement the following steps in Python.

   (a) Load the GT tower image. You can use the following command `X = np.loadtxt("tech_tower.txt")`. Look at what are loaded in the workspace.

   (b) Do an SVD decomposition on the X matrix.

   (c) Find the rank k approximation of X, for $k = 5, 15, 25$. Denote the rank k approximation of X as X_k.

   (d) Generate the image for X_k.

   You can use `plt.gray()`, `plt.imshow(X)`, and `plt.show()` to display the image from the matrix `X`. Besides, the package `numpy` also provides a quick command to do the SVD decomposition of a matrix.

   You should submit your code and the images.

   Solution: We plot the compressed images for $k = 5, 15, 25$ in Figures 2, 3, and 4, respectively.

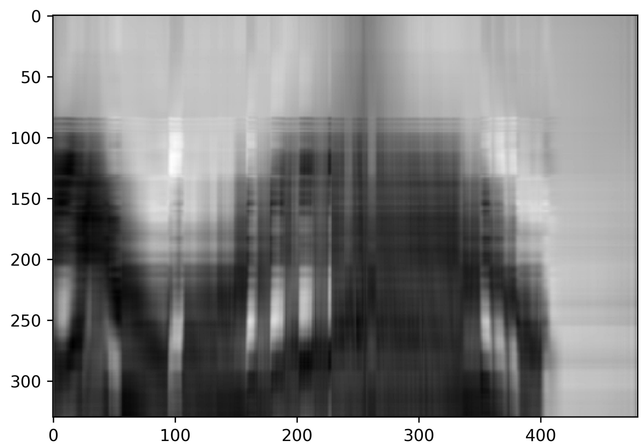

Figure 2: Compressed GT tower image with rank $k = 5$.

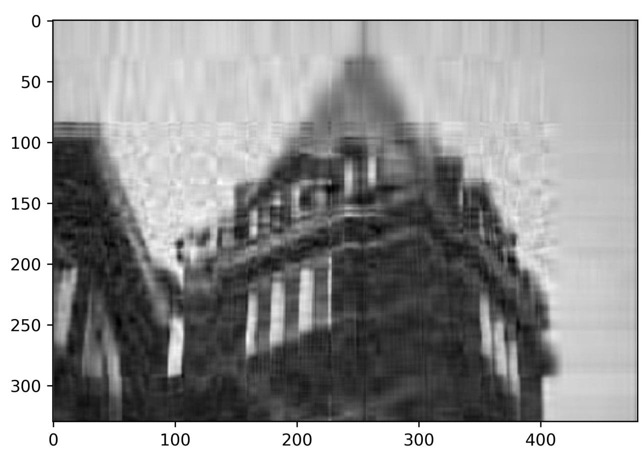

Figure 3: Compressed GT tower image with rank $k = 15$.

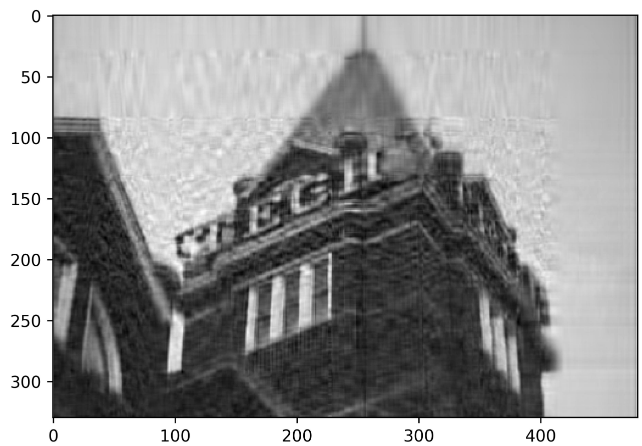

Figure 4: Compressed GT tower image with rank $k = 25$.

Python code:

```
import numpy as np
from matplotlib import pyplot as plt

X = np.loadtxt("texh_tower.txt")
plt.gray()

U, S, Vh = np.linalg.svd(X)
V = Vh.T

k = 5 # can be changed to 15, 25

Xk = U[:, 0: k].dot(np.diag(S[0: k], 0))
Xk = Xk.dot(V[:, 0: k].transpose())
plt.imshow(Xk)
plt.show()
```